

SHAILZA JOLLY

📍 Berlin, Germany

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PROFILE

Builder and researcher with a Ph.D. in CS and 6+ years spanning academic research and industry, working at the intersection of science, engineering, and product. Specializes in LLMs, Generative AI, and scalable data pipelines, with hands-on depth in modern LLM architectures, efficiency techniques (KV-cache, speculative decoding, FlashAttention), and agentic system design. Pairs a track record of top-tier publications (AAAI, NAACL, EMNLP) with shipping production systems and translating technical work into clear business outcomes.

SKILLS

- **ML/GenAI:** LLMs, NLU/NLG, multimodal learning, synthetic data, model evaluation.
- **LLM systems:** Agentic workflows (MCP, LangGraph/LangSmith), RAG/grounding concepts, experiment design & offline evaluation.
- **Data & scale:** Python, PyTorch, PySpark; large-scale text processing/data quality.
- **Search and retrieval:** embedding models (e.g., Sentence Transformers), reranking, vector search.
- **Platforms:** AWS; production-grade coding, documentation, and cross-functional delivery.

WORK EXPERIENCE

- Applied AI & LLM Systems**
Independent

July 2025 - Present
Berlin, Germany

 - Exploring problem spaces in LLM evaluation, enterprise AI, and agentic systems by building working prototypes: Sift, a hybrid retrieval system for technical documentation, and a personal AI learning tool built after finding the gap in existing apps.
 - Deepened LLM architecture fundamentals (Stanford) including efficiency internals: KV-cache, speculative decoding, FlashAttention.
- Flinn.ai**
Senior AI Engineer

January 2025 - June 2025
Berlin, Germany

 - Led development of AI-driven product features including data extraction from medical research papers and multilingual complaint monitoring.
 - Partnered with product and backend teams to scope problems, define success metrics, and deliver end-to-end features within an agile product lifecycle.
 - Navigated cost-performance tradeoffs under production constraints and learned to translate between engineers, product managers, and domain experts who each speak a fundamentally different language.
- Parental Leave (Maternity)**

December 2023 - December 2024

 - Watched a human brain acquire language and reasoning from nothing; the most grounding perspective on what AI is actually trying to replicate.
- Amazon AI**
Research Scientist

July 2022 - November 2023
Berlin, Germany

 - Led development of a scalable noise-removal/data-quality pipeline processing billions of tokens to improve training data for LLMs.
 - Presented findings and recommendations to senior science and engineering leadership; mentored Master's and Ph.D. interns on research deliverables.
- German Research Center for Artificial Intelligence (DFKI)**
Machine Learning Scientist

March 2019 - June 2022
Berlin, Germany

 - Implemented transformer-based ML systems end-to-end (GPU training, Docker, production-level code) within BMBF-funded projects; published results at AAAI, NAACL, and EMNLP.

- Collaborated with academic and industry partners internationally; supervised interns and BSc/MSc students.

NVIDIA Research

Machine Learning Scientist Intern

May - August 2021

Santa Clara, United States

- Built a document understanding pipeline combining table/cell detection, tabular structure retrieval, and OCR for financial documents.

Amazon Alexa

Applied Scientist Intern

August 2019 - December 2019

Aachen, Germany

- Developed a method for generating diverse synthetic training data to improve intent classification and slot labeling for task-oriented NLU.

SAP AI Research

Research Intern / Master's Thesis

May 2018 - February 2019

Berlin, Germany

- Designed and implemented an evaluation metric for Visual Question Answering (VQA) models.

EDUCATION

TU Kaiserslautern, Germany

Ph.D. in Computer Science

March 2019 - June 2022

Building Natural Language Generation and Understanding Systems in Data-Constrained Settings; work on VQA, interpretability, and conversational AI.

Overall grade: **“Sehr Gut” 1.0** (1.0 is highest on 1.0–5.0 scale)

University of Copenhagen, Denmark

Visiting Researcher

January 2021 - March 2021

Developed an unsupervised post-editing algorithm to generate fluent fact-checking explanations.

TU Kaiserslautern, Germany

M.Sc. in Computer Science — Minor in Economics

October 2016 - November 2018

Overall grade: **“Sehr Gut” 1.5**

Kyushu University, Japan

Semester Abroad

November 2017 - February 2018

Explainable AI to analyze behavior of deep CNN architectures for image recognition.

Guru Nanak Dev Engineering College, India

B.Tech in Computer Science & Engineering

August 2012 - July 2016

Overall grade: **First Division with Distinction**

SELECTED PUBLICATIONS

- **Jolly, S.**, Zhang, Z. X., Dengel, A., & Mou, L. (2022). Search and learn: Improving semantic coverage for data-to-text generation. *AAAI*.
- **Jolly, S.**, Pezzelle, S., & Nabi, M. (2021). EaSe: A diagnostic tool for VQA based on answer diversity. *NAACL-HLT*.
- **Jolly, S.**, & Kapoor, S. (2020). Can pre-training help VQA with lexical variations? *Findings of EMNLP*.
- **Jolly, S.**, Falke, T., Tirkaz, C., & Sorokin, D. (2020). Data-efficient paraphrase generation to bootstrap intent classification and slot labeling. *COLING (Industry Track)*.
- **Jolly, S.**, Iwana, B. K., Kuroki, R., & Uchida, S. (2018). How do convolutional neural networks learn design? *ICPR. (Best Student Paper)*

AWARDS AND HONORS

- Received **AAAI-22 Scholarship** by Hitachi to attend AAAI 2022 (2022).
- Awarded **AI Newcomer 2021 Award** (German Informatics Society and BMBF, Germany) (2021).
- Received **EU-Cost STSM Grant** to work on Multi3generation project at CopeNLU (2020).
- **Best Student Paper Award** at ICPR (2018).